

INDIAN INSTITUTE OF TECHNOLOGY KANPUR

Understanding the Value of a Firm

Prof. Abhinava Tripathi





Lesson 3: Making investment decisions

Module : Understanding the Value of a Firm

Introduction

In this lesson, we will cover the following topics:

- Review of NPV basics
- Alternatives to NPV rule – Payback period method
- Alternatives to NPV rule – Internal rate of return (IRR) method
- Pitfalls of IRR
- Capital investments with limited resources
- Summary and concluding remarks



Review of NPV Basics

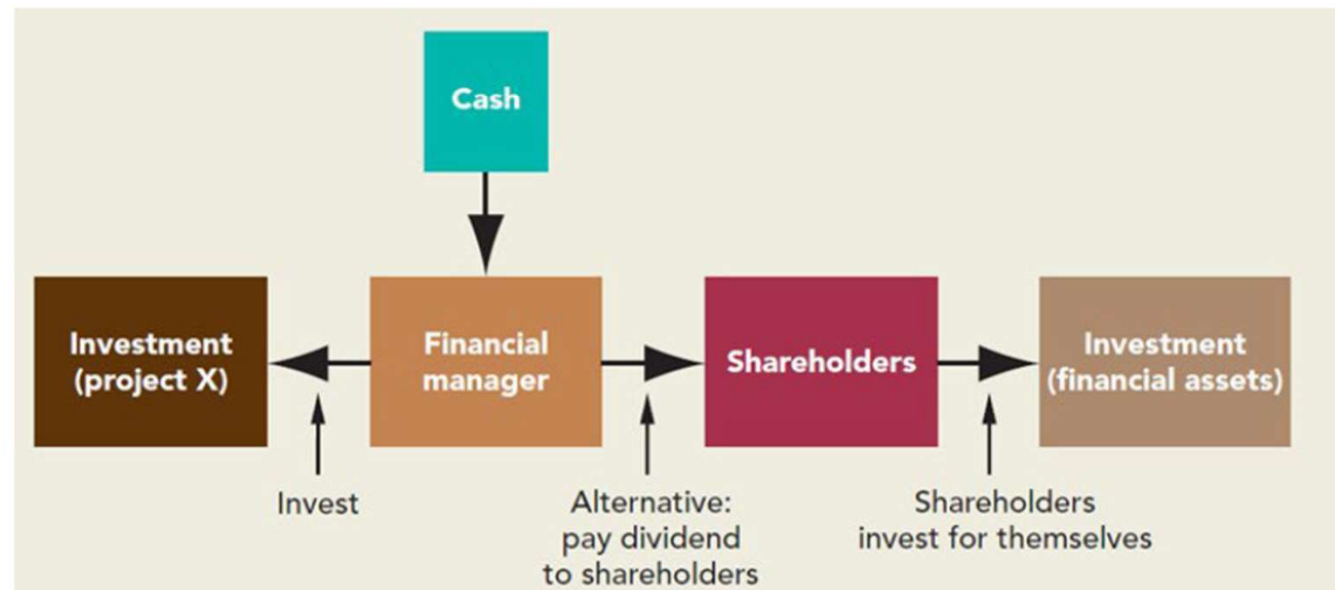
Review of NPV Basics

Consider yourself in a position of a CFO where you are analyzing \$1 million investment in a new venture called project P

- That the current market value of your firm is \$10 million, which includes \$1 million cash that you plan to invest in project P
- You find the NPV of this project by discounting the cash flows, adding them up to compute their PV, and subtracting the initial investment of \$1 million
- It is easy to understand if $PV > 1$ this project has a positive NPV

Review of NPV Basics

- Instead of investing in this project, you can distribute the cash to shareholders
- The project opportunity cost is precisely the rate of return that shareholders can obtain by investing in financial market instruments of similar risk instead of the project
- Assets of equivalent risks as your project should be chosen to estimate the opportunity cost or expected returns on your project



Brealey, Myers and Allen, *Principles of Corporate Finance*, 10th, 11th, or 12th editions, Chapter 5

Review of NPV Basics

The following are key aspects of NPV rule

- NPV rule recognizes that a dollar today is worth more than a dollar tomorrow and a safe dollar is worth more than a risky dollar
- Any decision rule that is affected by managers' tastes or choice of accounting method, the profitability of the existing business, or that of other projects will lead to an inefficient decision
- $NPV(A+B) = NPV(A) + NPV(B)$
- If project B has negative NPV, then $NPV(A+B)$, though positive, is lower than $NPV(A)$
- You would not take project B, just because it is packaged with a good project A

Review of NPV Basics

Companies report their book incomes frequently

- Book incomes are not necessarily the same as cash flows
- For example, depreciation is a non-cash expense, it is subtracted from book income to arrive at profits
- Profitability measures, such as book rate of returns, heavily depend on the classification of various items as capital investment and their rate of depreciation
- A project's selection or rejection should not depend on how accountants classify cash flows



Alternatives to NPV Rule – Payback Period Method

Alternatives to NPV Rule – Payback Period Method

A project's payback period is simply found by estimating the years it takes for the project cash flows to meet the initial investment

- A washing machine is costing \$800. You spend \$300 a year on washing your clothes. As a thumb rule, if this machine is purchased, it will recover its expenses in 3 years
- The payback rule states that a project should be accepted if its payback period is less than some cut-off period
- Consider a simple example here

Project	C0	C1	C2	C3	Payback Period (years)	NPV at 10%
A	-2,000	500	500	5,000	3	+2,624
B	-2,000	500	1,800	0	2	-58
C	-2,000	1,800	500	0	2	+50

Alternatives to NPV Rule – Payback Period Method

Cash flows corresponding to projects A, B, and C are provided, along with their payback period and NPVs at 10% discount rate

- The NPV rule suggests accepting projects A and C but rejects B
- Contrasting results are obtained as per the payback rule
- If a 2-year cut-off period is selected, then as per the payback rule, only projects B and C would be selected

Project	C0	C1	C2	C3	Payback Period (years)	NPV at 10%
A	-2,000	500	500	5,000	3	+2,624
B	-2,000	500	1,800	0	2	-58
C	-2,000	1,800	500	0	2	+50

Alternatives to NPV Rule – Payback Period Method

Thus payback rule offers misleading results for the following reasons

- It ignores the cash flows after the cut-off period
- It does not consider the time-value of money, and therefore, gives equal weight to all the cash flows before cut-off date
- If you entirely rely on payback rule, you will miss good projects with long-life and may accept poor projects that are short-lived

Project	C0	C1	C2	C3	Payback Period (years)	NPV at 10%
A	-2,000	500	500	5,000	3	+2,624
B	-2,000	500	1,800	0	2	-58
C	-2,000	1,800	500	0	2	+50

Alternatives to NPV Rule – Discounted Payback Period Method

An improved version of payback period is to employ discounted cash flows

- This discounted payback rule examines that how many years it takes for the discounted cash flows to recover the initial investment, i.e., become NPV positive
- This rule examines that the time it takes for the discounted cash flows to recover the initial investment

Project	C0	C1	C2	C3	Discounted Payback Period (years)	NPV at 10%
A	-2,000	$\frac{500}{1.1} = 455$	$\frac{500}{1.1^2} = 413$	$\frac{5,000}{1.1^3} = 3757$	3	+2,624
B	-2,000	$\frac{500}{1.1} = 455$	$\frac{1,800}{1.1^2} = 1488$	-	-	-58
C	-2,000	$\frac{1,800}{1.1} = 1636$	$\frac{500}{1.1^2} = 413$	-	2	+50



Alternatives to NPV Rule – Internal Rate of Return (IRR) Method

Alternatives to NPV Rule – Internal Rate of Return (IRR) Method

IRR rule comes from the simple return measure

- Project return = $\frac{\text{Profit}}{\text{Investment}} = \frac{\text{Payoff}}{\text{Investment}} - 1$; or $-\text{Investment} + \frac{\text{Payoff}}{1+\text{Project Return}} = 0$
- IRR is the return or discount rate at which NPV=0
- $NPV = C_0 + \frac{C_1}{(1+IRR)} + \frac{C_2}{(1+IRR)^2} + \dots + \frac{C_T}{(1+IRR)^T} = 0$

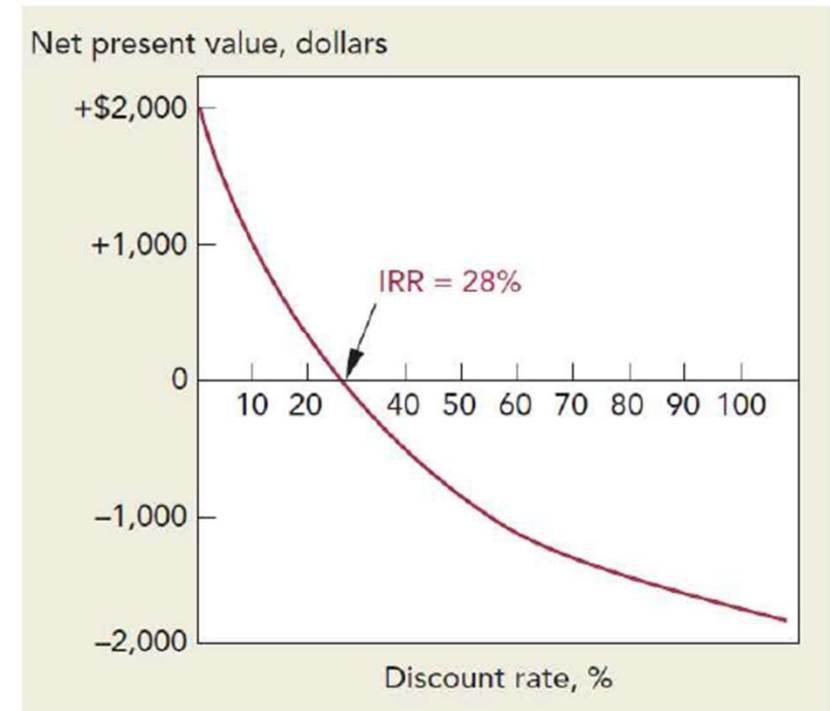
C_0	C_1	C_2
-4000	+2000	+4000

- $NPV = -4000 + \frac{2000}{1+IRR} + \frac{4000}{(1+IRR)^2} = 0$; solving for this, we get IRR= 28.08%

Alternatives to NPV Rule – Internal Rate of Return (IRR) Method

What happens to NPV as discount rate changes

- If the opportunity cost of capital is greater than the IRR, the project has a negative NPV
- If the opportunity cost is same as IRR, then $NPV=0$
- If opportunity cost of capital is smaller than the IRR, the project has a positive NPV
- We compare the opportunity cost of capital with the IRR on our project

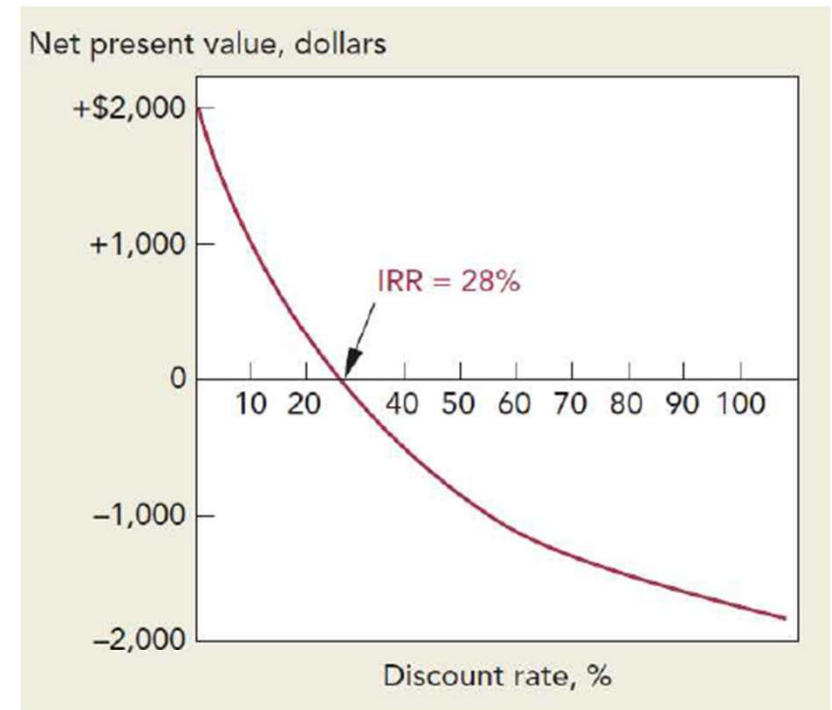


Brealey, Myers, and Allen, *Principles of Corporate Finance*, 10th, 11th, or 12th editions, Chapter 5

Alternatives to NPV Rule – Internal Rate of Return (IRR) Method

Often we tend to confuse opportunity cost with IRR

- IRR is a profitability measure and depends solely on the timing of the project cash flows
- The opportunity cost of capital is the standard of profitability to judge the worth (or NPV) of the project
- At the opportunity cost of capital, a project's NPV can be positive, negative, or zero

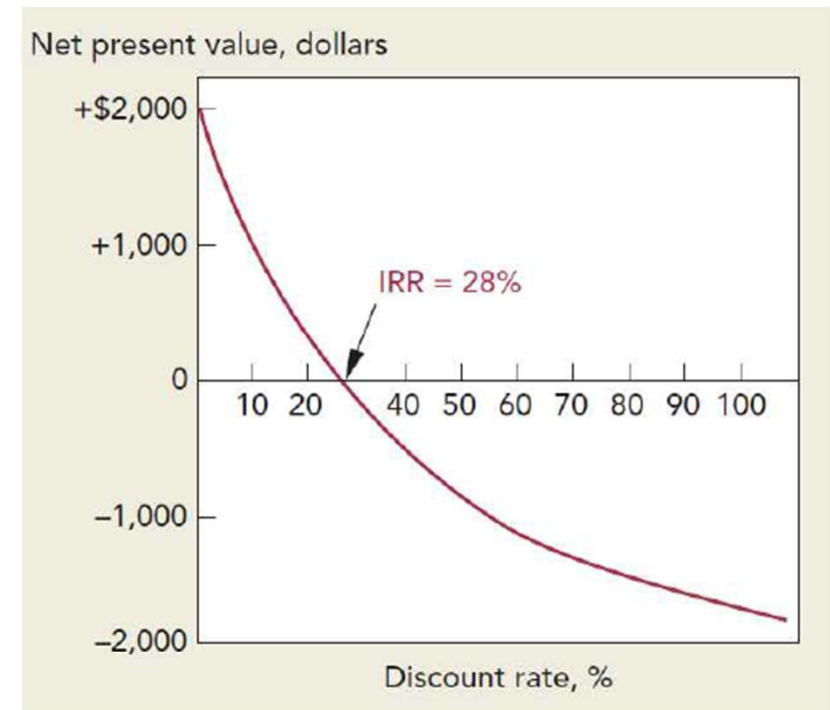


Brealey, Myers, and Allen, *Principles of Corporate Finance*, 10th, 11th, or 12th editions, Chapter 5

Alternatives to NPV Rule – Internal Rate of Return (IRR) Method

Often we tend to confuse opportunity cost with IRR

- If project cash flows are discounted at IRR, NPV is necessarily =0
- The opportunity cost is observed and estimated from capital markets by examining the securities with similar risk as that of the project
- IRR is intrinsic to the project, and depends on project cash flows only



Brealey, Myers, and Allen, *Principles of Corporate Finance*, 10th, 11th, or 12th editions, Chapter 5



Pitfalls of IRR

Pitfalls of IRR

Pitfall 1: Problem of Lending vs. borrowing

- Consider the project cash flows from projects A and B as shown here

Projects	C_0	C_1	IRR	NPV at 10%
A	-1000	+1500	50%	+364
B	1000	-1500	50%	-364

- Both of these projects will give you the same IRR
- In project A, we are paying out \$1000 initially, and getting \$1500 later - Case of lending
- While in case of B, we are initially getting \$1000 and paying back \$1500 later - Case of borrowing
- When you lend money, you want a higher return, and when you borrow money, you want a lower return

Pitfalls of IRR

Pitfall 1: Problem of Lending vs. borrowing

- For project B, you will find that NPV increases as discount rate increases

Projects	C_0	C_1	IRR	NPV at 10%
A	-1000	+1500	50%	+364
B	1000	-1500	50%	-364

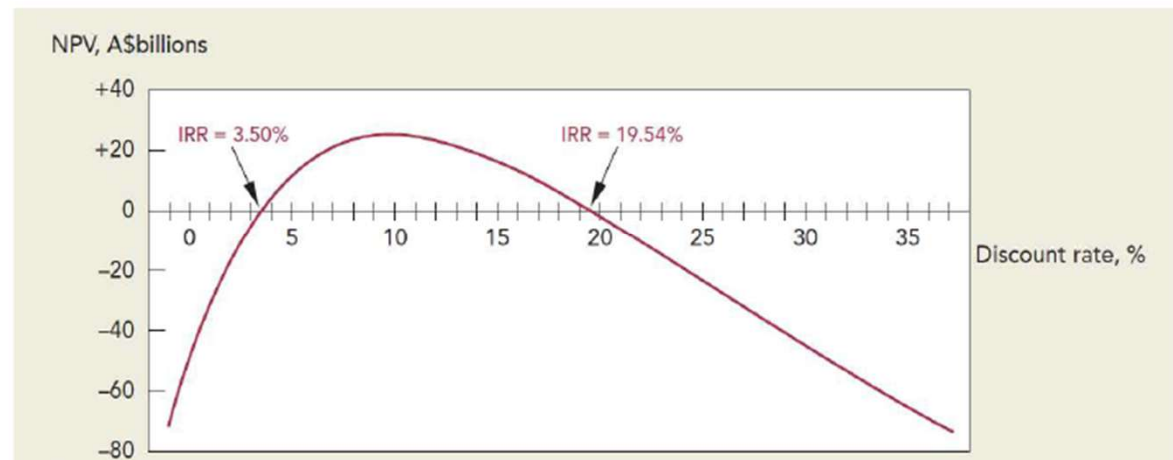
- For project B, you will find that NPV increases as discount rate increases
- The traditional way of looking at IRR will not work in this case, and we need to consider the rule in an opposite manner
- Select those opportunities of borrowing where IRR is less than the opportunity cost of capital

Pitfalls of IRR

Pitfall 2: Multiple rates of return

- Consider another project that involves an initial investment of \$3 billion and then produce a cash flow \$1 billion per year, for the next nine years
- At the end of the project, the company will incur \$6.5 billion of cleanup costs

C_0	C_1	C_2	C_3	C_4
-3	1	1	1	1
C_5	C_6	C_7	C_8	C_9
1	1	1	1	1

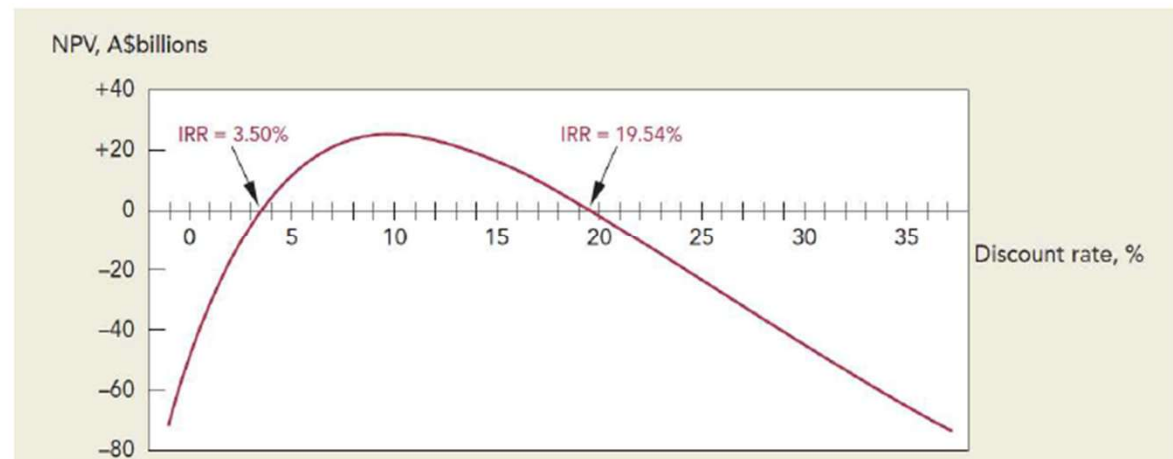


Pitfalls of IRR

Pitfall 2: Multiple rates of return

- There are two discount rates that make $NPV=0$
- As the discount rate rises, NPV initially rises and then declines, and crosses the zero NPV line two times

C_0	C_1	C_2	C_3	C_4
-3	1	1	1	1
C_5	C_6	C_7	C_8	C_9
1	1	1	1	1

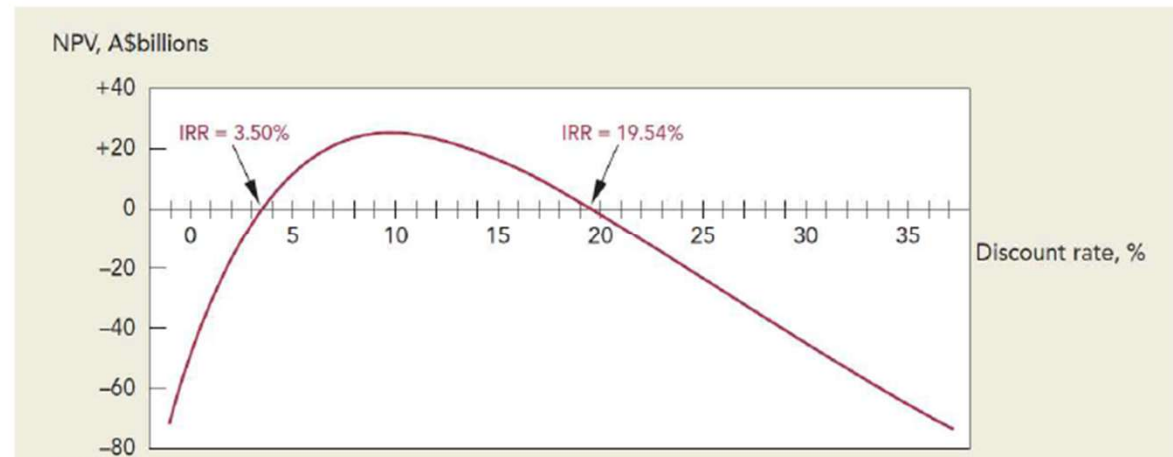


Pitfalls of IRR

Pitfall 2: Multiple rates of return

- The reason for this change in direction is change in cash flow signs
- So we can have as many IRRs as there are changes in the sign of cash flows

C_0	C_1	C_2	C_3	C_4
-3	1	1	1	1
C_5	C_6	C_7	C_8	C_9
1	1	1	1	1



Pitfalls of IRR

Pitfall 3: Mutually exclusive projects

- Firms often have to choose from mutually exclusive projects since it may not be feasible to take all of them
- Capital is a scarce resource, that is available only in limited quantities

Projects	C_0	C_1	IRR (%)	NPV at 10%
D	-10000	+20000	100	8182
E	-20000	+35000	75	11818

- Many times in such situations IRR can be misleading

Projects	C_0	C_1	IRR (%)	NPV at 10%
E-D	-10000	+15000	50	3636

Pitfalls of IRR

Pitfall 3: Mutually exclusive projects

- Consider the projects D and E here
- In the project cash flows shown here, it seems IRR and NPV are contradicting each other

Projects	C_0	C_1	IRR (%)	NPV at 10%
D	-10000	+20000	100	8182
E	-20000	+35000	75	11818

- The IRR rule suggests that project D is more profitable, NPV rule suggests that project E is more profitable

Projects	C_0	C_1	IRR (%)	NPV at 10%
E-D	-10000	+15000	50	3636

Pitfalls of IRR

Pitfall 3: Mutually exclusive projects

- In such cases, IRR can still be salvaged by examining incremental cash flows as shown here
- Compute the difference between E and D cash flows, examine the IRR on the additional \$10000 being

Projects	C_0	C_1	IRR (%)	NPV at 10%
D	-10000	+20000	100	8182
E	-20000	+35000	75	11818

- Do incremental cash flows from undertaking project D justify the investment

Projects	C_0	C_1	IRR (%)	NPV at 10%
E-D	-10000	+15000	50	3636

IRR in Conclusion

Many things can go wrong with IRR, but it is still a very useful benchmark

- IRR figure can convey the profitability of a project in the easiest manner
- To see its utility, have a look at the project cash flows, NPV, and IRR estimates for two projects X and Y as shown here (\$, thousands)

Projects	C_0	C_1	C_2	C_3	NPV at 8%	IRR (%)
X	-9.0	2.9	4.0	5.4	1.4	15.58
Y	-9000	2560	3540	4530	1.4	8.01

- Both of these projects offer the same positive NPV of \$1400
- As rational individuals, you would select X over Y (Why?)

IRR in Conclusion

It is an interesting example, as both of these projects offer the same positive NPV of \$1400

- As rational individuals, you would select X over Y (Why?)
- Investing in project X is clearly attractive as it offers you \$1400 by investing only \$9000 as against \$9 Mn (project Y)

Projects	C ₀	C ₁	C ₂	C ₃	NPV at 8%	IRR (%)
X	-9.0	2.9	4.0	5.4	1.4	15.58
Y	-9000	2560	3540	4530	1.4	8.01

- The higher IRR associated with X (15.58%) reflects the low risk and efforts involved as compared with Y
- Project Y is not probably worth the worry and time compared to project X



Capital Investments with Limited Resources

Capital Investments with Limited Resources

A firm increases its wealth if it accepts projects with positive NPV

- However, there are various constraints and limitations from undertaking all such projects
- Capital is a scarce resource and this becomes a problem of capital rationing
- When the capital is limited, firms need to select those projects that offer the highest NPV
- That is, maximum return for each dollar invested

Capital Investments with Limited Resources

Consider a simple problem of capital rationing shown here

- There are three projects on offer, A, B, and C and the firm has \$10 Mn to invest
- Thus, firms would like to select those projects that offer highest NPV per dollar of investment; the firm can raise maximum up to \$10 Mn in a year
- It can invest in project A or in projects B and C, but not in all three

Cash Flows (\$ Mn)					
Project	C0	C1	C2	NPV at 10%	
A	-10	+30	+5	21	
B	-5	+5	+20	16	
C	-5	+5	+15	12	

Capital Investments with Limited Resources

We focus on the projects where we get the maximum return for our investment

- That is, highest NPV project for the given investment
- This measure is often called the profitability index
- Profitability index (PI) = $\frac{\text{NPV}}{\text{Initial Investment}}$

Cash Flows (\$ Mn)					
Project	C0	C1	C2	NPV at 10%	
A	-10	+30	+5	21	
B	-5	+5	+20	16	
C	-5	+5	+15	12	

Capital Investments with Limited Resources

We focus on the projects where we get the maximum return for our investment

- As per the PI criteria, you would first select project B and then C
- If the budget limit is \$10 Mn, we should accept only these two projects
- However, real life examples are not so simple

Cash Flows (\$ Mn)					
Project	C0	C1	C2	NPV at 10%	PI
A	-10	+30	+5	21	2.1
B	-5	+5	+20	16	3.2
C	-5	+5	+15	12	2.4

Capital Investments with Limited Resources

- Let us add another project D, which needs \$40 Mn investment in second year

Project	C0	C1	C2	NPV at 10%	PI
A	-10	+30	+5	21	2.1
B	-5	+5	+20	16	3.2
C	-5	+5	+15	12	2.4
D	0	-40	+60	13	0.4

- The firm can only raise \$10 Mn in the second year: additional constraint of capital rationing
- The simple way of ranking projects as per PI may not work here
- This particular problem is rather simple, as A and D combined offers a higher NPV than B and C combined
- However, more complex problems are solved with linear programming (LP) techniques



Summary and Concluding Remarks

Summary and Concluding Remarks

- In addition to NPV, other rules are also employed to examine alternate investments
- These include book rate of return, payback period, and IRR method
- Book rate of return is simply computed as book income divided by book value of investment
- Payback method examines the project cash flows against a certain specific cut-off period
- Only those projects with payback period less than cut-off period are considered
- Lastly, IRR is the discount rate at which the firm NPV is zero
- As per the IRR rule, firms should accept those projects that have an IRR greater than opportunity cost of capital

Thanks!